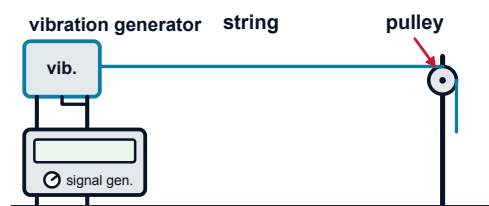


Waves on a string

A student measures the speed of a wave on a stretched string.

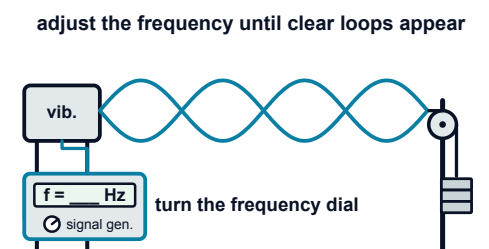
- 1** Attach one end of a string to a vibration generator and pass the other end over a pulley.



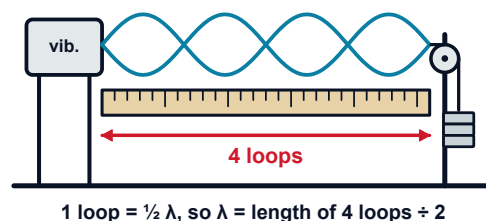
- 2** Hang masses on the end of the string to keep it taut.



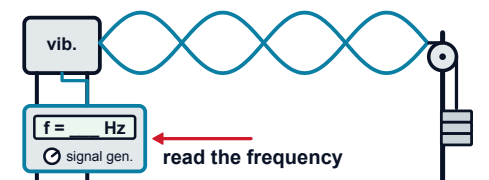
- 3** Switch on the signal generator and adjust the frequency until clear loops appear.



- 4** Measure the length of several loops and work out the wavelength.



- 5** Read the frequency from the signal generator.



- 6** Calculate the wave speed: frequency × wavelength.

$$v = f \times \lambda$$

wave speed = frequency × wavelength
m/s = Hz × m

use f from step 5 and λ from step 4

Note: The frequency can be read before or after the wavelength is measured.